Feature Fusion-Guided Network With Sparse Prior Constraints for Unsupervised Hyperspectral Image Quality Improvement

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***Abstract*— Due to imaging hardware limitations and atmospheric interference, hyperspectral image (HSI) often suffers from low spatial resolution or mixed noise degradation. HSI fusion and denoising are two key strategies to improve HSI quality. Traditional model-based methods rely on data-specific manual priors, while supervised deep learning methods are typically developed specifically for a single task and require a large number of training datasets. To address these limitations, we propose a novel unsupervised feature fusion-guided network (UFFGNet) as a general prior that effectively leverages multiscale semantic features from a guidance image while incorporating sparse prior constraints to suppress outliers. Specifically, UFFGNet comprises a deep feature extraction (DFE) network to capture multiscale semantic features from a guidance image and an attention-based feature generation (AFG) network that generates an output image from random noise. These two networks are connected by a feature refinement module (FRM) to embed the refined features from the feature aggregation module (FAM) into the generation network. Furthermore, the sparse prior constraint is incorporated to model sparse noise (including impulse noise, stripe artifacts, and deadlines) in HSI, thereby improving the robustness of UFFGNet. The proposed network optimizes network parameters in an unsupervised manner without external additional training data, using the fidelity term in the degradation model as a loss function to learn the prior information of the original image. Extensive experiments demonstrate that the proposed UFFGNet outperforms state-of-the-art methods in both HSI fusion and denoising tasks, significantly improving HSI quality.**

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***Index Terms*— Deep image prior (DIP), HSI denoising, HSI fusion, hyperspectral image (HSI), unsupervised learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) are acquired across a continuous wavelength band, providing detailed spectral and spatial information about areas of interest. Benefiting from the rich information, HSIs are widely used in various applications, such as object detection [1], classification [2], and unmixing [3]. However, due to physical constraints of the sensor or complex illumination conditions, HSIs often experience various forms of degradation, including low spatial resolution and mixed noise. To improve HSI quality, HSI fusion and denoising are two key strategies. HSI fusion aims to reconstruct the high-spatial-resolution HSI (HR-HSI) through the fusion of a low-spatial-resolution HSI (LR-HSI) and a high-spatial-resolution multispectral image (HR-MSI), while HSI denoising aims to restore the original clean HSI from degraded noisy HSI (see Fig. 1).

In general, current HSI fusion and denoising approaches can be roughly categorized into traditional model-based methods and deep-learning-based (DL-based) methods. Traditional model-based methods aim to directly formulate an optimization model by considering proper prior regularizers and fidelity terms to obtain the desired HSI. The common priors in HSI fusion or denoising include low-rank [4], [5], [6], [7], [8], [9], [10], sparsity [11], total variation [12], [13], [14], [15], and nonlocal similarity [16], [17]. However, these handcrafted priors are designed for specific datasets and need to adjust the regularization parameters manually. DL-based methods [18], [19], [20], [21], [22], [23], [24], [25] leverage powerful neural network models to exploit the implicit priors of HSI, with each model tailored to a specific task and using either supervised or unsupervised learning. The objective of supervised learning methods is to learn a comprehensive mapping function from abundant pairs of training data [24], [25], [26], [27], [28]. Due to the inconsistency of training datasets across different tasks, the trained models are limited to specific tasks. Recently, unsupervised learning approaches have been successfully used for various problems without any training data [29], [30], [31], [32], [33], [34], [35]. However, these unsupervised learning methods primarily leverage implicit priors derived from testing data, failing to fully use the semantic features or detailed information provided by guidance data. In summary, although

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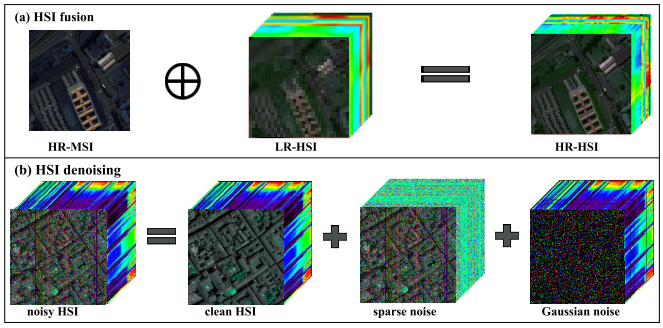


Fig. 1. Schematic illustrations of two HSI processing tasks that can significantly enhance the quality of HSI. (a) HSI fusion, which involves reconstructing the HR-HSI from HR-MSI and LR-HSI. (b) HSI denoising, which involves restoring the original clean HSI from degraded noisy HSI.

significant progress has been made in HSI fusion and denoising methods, the existing approaches are applicable only to either specific datasets or tasks, resulting in a lack of flexibility in practical applications. Moreover, they overlook the rich spatial context and detailed features embedded in the guidance image. Furthermore, these methods often lack robustness and struggle to address outliers effectively during the image degradation process.

To address the aforementioned challenges, we propose a novel unsupervised feature fusion-guided network (UFFGNet) for HSI fusion or denoising. The motivation behind the proposed method is based on the availability of high-quality data in HSI fusion or denoising tasks, which can be effectively used to guide the generation of desired HSI. Furthermore, to effectively handle outliers, we extend UFFGNet to incorporate robust unsupervised learning by accounting for sparse components such as impulse noise, stripe artifacts, and deadlines. The proposed UFFGNet consists of two networks: a deep feature extraction (DFE) network and an attention-based feature generation (AFG) network. The DFE network is devised to extract multiscale encoding and decoding fusion features from a guidance image, whereas the AFG network focuses on generating a desired image from random noise. The DFE and AFG networks are connected through a feature refinement module (FRM), which embeds the fusion features into the generation network. The main contributions of this work can be summarized as follows.

1) We propose a novel UFFGNet that leverages multiscale features from a guidance image to generate high-quality HSI while incorporating sparse prior constraints to enhance learning robustness. The network parameters are optimized in an unsupervised manner without relying on external training datasets, improving flexibility and adaptability.
2) We design a feature aggregation module (FAM) to fuse the encoding and decoding features of guidance images, which can preserve their complementary structure characteristics. Moreover, an AFG is constructed to effectively leverage multiscale guidance information from the FAM, thereby enhancing the quality of HSI generation.
3) We apply the proposed unsupervised network to two representative HSI quality improvement tasks: HSI fusion and mixed noise removal. The experimental results from both the simulated and real data demonstrate that UFFGNet outperforms state-of-the-art methods in HSI fusion and mixed noise removal.

II. Related Work

This section provides an overview of the HSI fusion and denoising tasks, which are crucial for enhancing HSI quality.

A. HSI Fusion

Due to the significant tradeoffs between the spatial and spectral resolutions of the traditional HSI equipment, HSI typically has high spectral resolution but lower spatial resolution. On the contrary, RGB or multispectral sensors can easily collect higher spatial resolution images but at the cost of lower spectral resolution. HSI fusion aims to obtain HR-HSI by fusing LR-HSI and HR-MSI. The existing methods can be broadly divided into three categories: traditional model-based methods, DL-based methods, and guidance-based methods.

1) Traditional Model-Based Methods: Traditional model-based methods establish mathematical models to describe the relationship among the LR-HSI, HR-MSI, and HR-HSI, leveraging prior information about the desired HR-HSI. Total variation regularization [36] enforces piecewise smoothness, while sparse representation techniques [11], [37] model image sparsity through dictionary construction for HSI fusion. Low-rank matrix methods [4], [38], [39] have been used to capture contextual information in high-dimensional data for HSI fusion. However, since matrix factorization destroys the spatial correlation of HR-HSI, various tensor factorization techniques [40], [41], [42], [43] have been used to better preserve spatial relationships in HSI fusion. Although traditional model-based methods have shown effectiveness, they often struggle to effectively represent complex data scenarios.

2) DL-Based Methods: With the advancement of deep learning, numerous DL-based methods have been proposed. Depending on whether training datasets are required, DL-based methods are primarily categorized into supervised learning methods and unsupervised learning methods. Supervised learning methods [44], [45], [46] learn the end-to-end mapping from LR-HSI and HR-MSI to HR-HSI. For instance, convolutional neural-network-based approaches such as PzResNet [47] and PanNet [48] have improved the performance of HSI fusion by extracting local features through convolutional operations, while transformer-based networks such as Fusformer [49], Interactformer [46], and Essaformer [24] have attracted significant interest due to their excellence in modeling long-range dependencies for HSI fusion. However, these supervised learning methods typically require large amounts of training data and need to be retrained for different scale factors, limiting their generalization ability. To alleviate this issue, unsupervised learning methods, such as the sparse Dirichlet-net [31], model-guided coarse-to-fine fusion network [50], and adversarial autoaugmentation network [32], have been developed to learn the implicit prior of HR-HSI from testing data. However, these methods do not

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fully leverage the rich spatial context and intricate features present in the HR-MSI, which may lead to suboptimal performance.

3) Guidance-Based Methods: Guidance-based approaches [51] improve HSI fusion performance by designing a neural network that effectively leverages the spatial details of HR-MSI or the spectral characteristics of LR-HSI. GDD [52] proposed a novel unsupervised deep learning method that can be applied to image denoising and image fusion without the need for training. GuidedNet [53] proposed a general CNN fusion framework, which has been successfully applied to multiple image resolution enhancement problems. CS2DIPs [54] introduced nonnegative matrix–vector tensor factorization into the deep image prior (DIP) [29] framework to jointly learn the abundance tensor and spectral feature matrix from HR-MSI and LR-HSI, respectively. However, these methods either require large datasets for training or lack robustness, making it challenging to effectively address outliers.

B. HSI Denoising

Due to physical mechanisms, sensor sensitivity, and imaging conditions, HSI is inevitably polluted by various kinds of noise, such as Gaussian noise and sparse noise (stripe noise, deadline noise, and impulse noise). HSI denoising aims to restore a clean HSI from a noisy observation and can be categorized into three types: traditional model-based methods, DL-based methods, and guidance-based methods.

1) Traditional Model-Based Methods: Traditional model-based methods designed handcrafted priors, such as local smoothness [13], [14], [15], low-rank matrix approximation [55], [56], low-rank tensor factorization [57], [58], and nonlocal similarity [59], to regularize the restoration process. Zhang et al. [5] first proposed low-rank matrix approximation for HSI mixed noise removal. He et al. [55] integrated local smoothness and low-rank matrix approximation to improve denoising performance. Wang et al. [57] used low-rank tensor factorization and total variation to capture spatial–spectral correlation and local smoothness. He et al. [59] introduced a unified paradigm framework that concurrently captures global spectral correlation and the spatial nonlocal similarity within HSI. However, these methods lack the adaptive capability to capture the nonlinear feature distribution of HSI. Furthermore, they are burdened by high computational costs due to time-consuming optimization processes.

2) DL-Based Methods: Unlike traditional model-based methods, DL-based methods leverage the deep network's powerful ability for nonlinear feature extraction, substantially enhancing denoising outcomes. Supervised learning methods [60], [61] leverage extensive datasets to learn end-to-end mappings from inputs to outputs. Convolutional neural-network-based approaches, such as HSI-DeNet [62], effectively enhance image restoration by extracting local features via convolutional operations. Transformer-based methods such as SST [27] and SERT [63] have shown strong potential in modeling long-range dependencies. In addition, Mamba-based methods, including SSUMamba [64], have gained attention for their efficient long-range dependency modeling and computational effectiveness. Due to the complex distribution of noise in real-world scenarios, supervised learning methods may not be applicable to different scenes. Unsupervised learning methods [65], which do not rely on training datasets, are gradually being widely applied in HSI denoising. Miao et al. [66] introduced an unsupervised DIP framework, grounded in the classic spatiospectral decomposition of HSI. However, these methods fail to fully exploit the rich spatial context and detailed features embedded in the guidance image, resulting in suboptimal solutions.

3) Guidance-Based Methods: Guidance-based methods have gained significant attention in HSI denoising by leveraging high signal-to-noise ratio bands (HSNRBs) or auxiliary PAN images as complementary information to enhance denoising performance. GDD [52] introduced a novel network architecture that acts as a regularizer for noisy no-flash image denoising, guided by flash images. SGNLR [67] proposed a low-rank subspace representation that leverages global spectral correlation in HSI, reformulating the denoising task as the estimation of spectral basis and spatial coefficients, guided by HSNRBs. Pan-Denoising [68] introduced a novel paradigm for HSI denoising, guided by PAN images. However, the existing approaches are often limited to specific datasets or tasks, resulting in a lack of flexibility for practical applications.

III. Proposed Method

In this section, we present the proposed UFFGNet in detail. We first discuss the challenges of the existing methods for HSI fusion or denoising. Subsequently, we introduce the framework of the proposed UFFGNet in Fig. 2. Finally, we present the application of the proposed UFFGNet in two representative HSI tasks: fusion and denoising.

As introduced in the related work section, the existing approaches often face limitations, being applicable only to either specific datasets or tasks, which results in a lack of flexibility in practical applications, or failing to leverage the rich spatial and spectral features in the guidance image. Furthermore, these methods often lack robustness and face challenges in effectively handling outliers. Therefore, effective and robust unsupervised learning methods are still highly desired, as researchers aim to integrate HSI fusion and denoising under a unified framework. Unfortunately, most existing algorithms cannot fulfill this basic requirement. This inspires us to propose an unsupervised learning method that does not rely on training datasets, can fully use guidance information from testing data, and is applicable to HSI fusion or denoising.

A. Network Architecture

As depicted in Fig. 2, the proposed UFFGNet consists of two main components: a DFE network and an AFG network. These networks are connected via an FRM, specifically designed to effectively exploit and incorporate multiscale spatial–spectral information extracted from the guidance image into the AFG.

1) DFE Network: The DFE is similar to the U-Net and uses an encoder–decoder network to extract the multiscale hierarchical features of the guidance image from low-level

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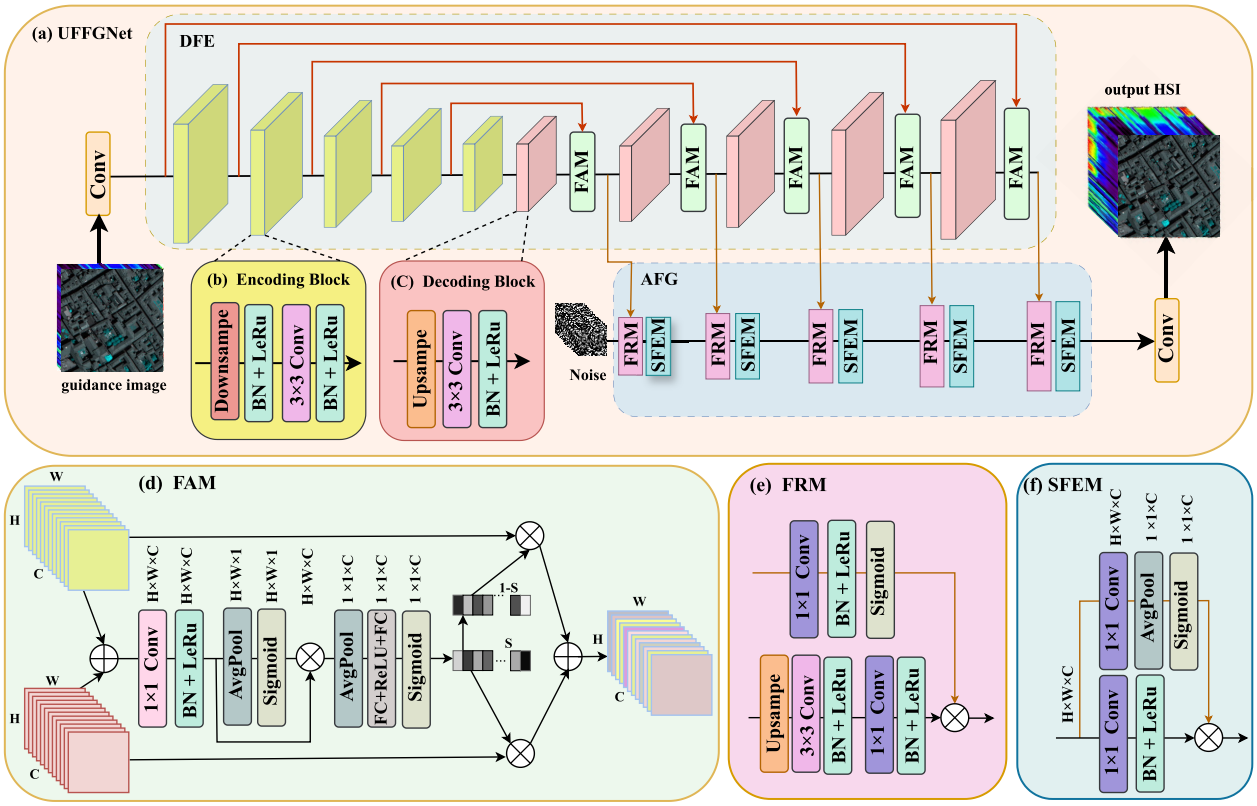


Fig. 2. Structure of the proposed UFFGNet. The multiscale semantic features are extracted from the guidance image by the DFE network. The semantic features are then used to generate an output image by the FRM and SFEM in the AFG network. (a) UFFGNet. (b) Encoding block. (c) Decoding block. (d) FAM. (e) FRM. (f) SFEM.

to high-level. First, the feature F_{en}^0 is produced by a convolution operation that transforms the guidance image into a feature map. Subsequently, multiple encoding blocks are used to obtain the multiscale features F_{en}^k. Similarly, the decoding features F_{dc}^k can be generated using the decoding block that is paired with the encoded features across the spatial-channel dimension. Notably, most of the commonly used techniques for feature fusion in U-Net encoding and decoding involve simple concatenation or summation combined with a skip connection. However, these operations make it challenging to depict the interaction of encoded and decoded features, resulting in insufficient feature fusion. In addition, the convolution operation in the encoding and decoding block has a local receptive field thus being unable to leverage contextual information outside the receptive field. To overcome the limitations of the existing U-Net structures in encoding and decoding feature fusion, we propose an FAM to mitigate information loss in the encoding–decoding process and additionally preserve their complementary semantic characteristics. Specifically, the FAM can dynamically aggregate the encoding and decoding features through spatial and channel attention mechanisms. The FAM first fuses the encoding and decoding features by elementwise sum and convolution operations and then performs global average pooling (GAP) over both the spatial and channel dimensions to generate statistical information. The spatial statistical information is used to generate spatial weights through the sigmoid function, effectively capturing spatial correlations. Meanwhile, channel statistical information is used to compute the weight vector via a multilayer perceptron and a sigmoid function. The overall FAM process is defined as

$$F_{\text{fam}}^k = \begin{cases} f_{\text{fam}}^1\left(F_{\text{dc}}^1, f_{\text{skip}}\left(F_{\text{en}}^{K-1}\right)\right), & k = 1 \\ f_{\text{fam}}^k\left(F_{\text{dc}}^{k-1}, f_{\text{skip}}\left(F_{\text{en}}^{K-k}\right)\right), & 1 < k < K \end{cases} \tag{1}$$

where $f_{\text{skip}}^k(\cdot)$ represents the extraction of information from the kth encoded features, $F_{\text{fam}}^k(\cdot)$ denotes the features generated by the FAM, and $f_{\text{fam}}^k(\cdot)$ denotes the feature aggregation process within the FAM. In summary, the DFE network plays a crucial role in extracting multiscale spatial–spectral features, which serve as essential guidance for feature generation.

2) AFG Network: The AFG network consists of FRM and squeeze feature excitation module (SFEM), where the FRM is designed to use abundant multiscale spatial–spectral information from the guidance image and the SFEM is used to efficiently exploit channel dependencies. Since we use the FAM to aggregate the encoded and decoded features, $F_{\text{fam}}^k(\cdot)$ contains more fine-grained characteristics and abundant semantic information. In the FRM, we first generate attention weights using fine-grained features $F_{fam}^k(\cdot)$, and then use these weights to guide semantic feature generation. In conclusion, the operation of the FRM can be described as follows:

$$F_{\text{frm}}^k = \begin{cases} f_{\text{frm}}^1\left(\mathcal{E}, F_{\text{fam}}^1\right), & k = 1 \\ f_{\text{frm}}^k\left(F_{\text{sfem}}^{k-1}, F_{\text{fam}}^k\right), & 1 < k < K \end{cases} \tag{2}$$

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where $F^k_{\text{frm}}(\cdot)$ and $F^k_{\text{sfem}}(\cdot)$ denote the features generated by the FRM and SFEM, respectively, $\mathcal{E}$ represents random noise, and $f^k_{\text{frm}}(\cdot)$ denotes the FRM operation in the kth layer.

To enhance channel dependency, we further design an SFEM to enhance the feature representational capacity by enabling dynamic channelwise feature recalibration. The SFEM can compress global spatial information into a channel descriptor, effectively capturing critical details while filtering out irrelevant ones. The operation of the SEFM module can be described as follows:

$$F^k_{\text{sfem}} = f^k_{\text{sfem}}\left(F^k_{\text{frm}}\right) \tag{3}$$

where $f^k_{\text{sfem}}(\cdot)$ represents SEFM operation in the kth layer. Finally, a simple convolution operation is applied to transform the enriched features extracted from the previous operations into an output image.

B. Application to HSI Fusion

HSI fusion aims to generate a high-quality HR-HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$ from a low-quality LR-HSI $\mathcal{Y} \in \mathbb{R}^{h\times w\times B}$ and an HR-MSI $\mathcal{Z} \in \mathbb{R}^{H\times W\times b}$, where $H \gg h$, $W \gg w$, and $B \gg b$. The degradation process between the HR-HSI and LR-HSI/HR-MSI can be expressed as follows:

$$\mathcal{Y} = \mathcal{X} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2, \quad \mathcal{Z} = \mathcal{X} \times_3 \mathbf{U}_3 \tag{4}$$

where $\mathbf{U}_1 \in \mathbb{R}^{h\times H}$, and $\mathbf{U}_2 \in \mathbb{R}^{w\times W}$ are the degradation matrix of blurring and downsampling in spatial height and width modes, respectively. $\mathbf{U}_3 \in \mathbb{R}^{b\times B}$ is the spectral response matrix. $\times_k$ is the mode-k tensor-matrix product, and the product of a tensor $\mathcal{M} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_n}$ and a matrix $\mathbf{V} \in \mathbb{R}^{J\times I_k}$ yields an $I_1 \times I_2 \times \cdots \times I_{k-1} \times J \times I_{k+1} \cdots \times I_n$ tensor denoted by $\mathcal{M} \times_k \mathbf{V}$ and satisfies $(\mathcal{M} \times_k \mathbf{V})_{i_1\cdots i_{k-1} j i_{k+1}\cdots i_n} = \sum_{i_k=1}^{I_k} m_{i_1 i_2\cdots i_n} v_{j i_k}$.

In general, the problem of generating HR-HSI based on an unsupervised method can be formulated as the following optimization problem:

$$\min_{\mathcal{X}} \|\mathcal{Y} - \mathcal{X} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2\|_F^2 + \lambda\|\mathcal{Z} - \mathcal{X} \times_3 \mathbf{U}_3\|_F^2 + R(\mathcal{X}) \tag{5}$$

where the first two terms are the model fidelity term, $R(\mathcal{X})$ is the regularization term describing prior information about the HR-HSI, and λ is the parameter used to balance these terms. Traditional model-based methods exploit handcrafted priors by designing a corresponding regularization term, denoted as $R(\mathcal{X})$. However, the task-specific prior cannot be easily extended to other tasks. Instead of using the handcrafted prior, we use the proposed UFFGNet to estimate $\mathcal{X}$ by adaptively learning the deep prior of HR-HSI as follows:

$$\mathcal{X} = \mathbf{T}_\Theta(\mathcal{G}, \mathcal{E}) \tag{6}$$

where $\mathbf{T}_\Theta(\mathcal{G}, \mathcal{E})$ denotes UFFGNet with the network parameters Θ, and $\mathcal{G}$ and $\mathcal{E}$ are the network inputs representing the guidance image and random noise, respectively. Due to the consistent spatial resolution between HR-MSI and HR-HSI, we can use HR-MSI as the guidance image input to the network to generate HR-HSI, i.e., $\mathcal{G} = \mathcal{Z}$. Therefore, the optimization problem (5) can be rewritten as follows:

$$\min_{\Theta} \|\mathcal{Y} - \mathbf{T}_\Theta(\mathcal{G}, \mathcal{E}) \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2\|_F^2 + \lambda\|\mathcal{Z} - \mathbf{T}_\Theta(\mathcal{G}, \mathcal{E}) \times_3 \mathbf{U}_3\|_F^2. \tag{7}$$

It is worth noting that the optimization of parameter Θ is a regression problem, and the Adam algorithm [69] is used as the optimizer in the proposed UFFGNet. The loss function of UFFGNet is defined as an optimization problem (7). The first term constrains the spectral consistency between the spatially degraded $\mathcal{X}$ and $\mathcal{Y}$. In contrast, the second term constrains the spatial consistency between the spectral degraded $\mathcal{X}$ and $\mathcal{Z}$. During the optimization of network parameters, only random noise and a known guidance image are used as inputs, with the loss function excluding any reliance on external data. The network parameters are dynamically optimized during the testing phase using a carefully designed loss function, relying solely on the provided input degradation data. The gradient of the parameter Θ in UFFGNet is computed using the standard backpropagation method. As a result, the proposed method eliminates the need for any training dataset, establishing it as an unsupervised deep learning framework.

C. Application to HSI Denoising

HSI is inevitably contaminated by various kinds of noise, such as Gaussian noise, stripe noise, deadline noise, and impulse noise. To enhance the adaptability of UFFGNet in handling these complex restoration challenges, we further extend our UFFGNet to robust unsupervised restoration. HSI denoising aims to restore the desired HSI $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$ from observed noisy HSI $\mathcal{Y} \in \mathbb{R}^{H\times W\times B}$. The degradation model of noisy HSI can be formulated as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N} \tag{8}$$

where $\mathcal{S} \in \mathbb{R}^{H\times W\times B}$ and $\mathcal{N} \in \mathbb{R}^{H\times W\times B}$ denote sparse noise (including stripe noise, deadline noise, and impulse noise) and Gaussian noise, respectively.

Analogously, the proposed UFFGNet is designed to learn the deep prior and estimate the desired HSI $\mathcal{X}$. To eliminate the influence of sparse noise, we propose to use a robust restoration model as follows:

$$\min_{\Theta,\mathcal{S}} \|\mathcal{Y} - \mathbf{T}_\Theta(\mathcal{G}, \mathcal{E}) - \mathcal{S}\|_F^2 + \lambda\|\mathcal{S}\|_1 \tag{9}$$

where $\|\mathcal{S}\|_1$ is the sparse regularization of sparse noise, which is defined as the sum of the absolute values of all the elements. Since the noisy HSI contains many HSNRBs [67], we use these high-quality bands as guidance image $\mathcal{G}$ for the UFFGNet, which can fully use the information in the noisy data.

There are two variables to optimize in the model (9), and we use an effective alternating minimization to optimize it. Fixing variable $\mathcal{S}$, Θ subproblem is formulated as

$$\min_{\Theta} \|\mathcal{Y} - \mathbf{T}_\Theta(\mathcal{G}, \mathcal{E}) - \mathcal{S}\|_F^2 \tag{10}$$

which also uses the Adam algorithm as the optimizer in the proposed UFFGNet, and the loss function is set as model (10).

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TABLE I
COMPARATIVE QUANTITATIVE ANALYSIS OF VARIOUS ALGORITHMS IN SIMULATED HSI FUSION EXPERIMENTS

Dataset	Scene	Index	HySure [36]	FSTRD [43]	GTNN [7]	CTDF [8]	GDD [52]	UMC2FF [50]	ADASR [32]	CS2DIP [54]	uHNTC [51]	UFFGNet
CAVE	Balloons	PSNR	36.67	39.16	34.00	43.24	47.14	43.89	44.08	43.62	44.60	**47.98**
		SSIM	0.968	0.979	0.970	0.993	0.996	0.995	0.995	0.993	0.995	**0.997**
		SAM	9.88	8.07	4.69	3.19	2.19	2.80	2.72	3.39	2.73	**1.93**
	Beads	PSNR	29.47	30.72	28.38	30.46	35.35	31.29	25.84	34.30	33.28	**37.08**
		SSIM	0.851	0.899	0.886	0.917	0.973	0.948	0.886	0.970	0.963	**0.974**
		SAM	16.64	16.79	12.01	11.75	5.86	7.53	9.82	6.44	7.54	**5.55**
	Cd	PSNR	30.36	29.44	30.38	32.10	34.76	31.97	30.55	33.62	34.78	**35.23**
		SSIM	0.931	0.923	0.949	0.968	0.979	0.972	0.970	0.972	0.978	**0.984**
		SAM	18.77	20.53	6.97	5.23	4.01	5.32	4.71	5.17	4.60	**3.45**
	Toy	PSNR	34.86	36.84	32.33	37.15	42.70	38.50	35.71	40.12	39.76	**43.72**
		SSIM	0.949	0.961	0.959	0.966	0.983	0.984	0.981	0.988	0.988	**0.993**
		SAM	16.00	14.20	8.38	8.79	5.34	6.83	6.68	6.01	4.98	**4.34**
	Beers	PSNR	36.34	42.37	34.63	42.68	43.57	39.59	39.05	42.60	42.12	**44.72**
		SSIM	0.978	0.980	0.969	0.992	0.992	0.989	0.987	0.992	0.991	**0.993**
		SAM	3.27	12.05	2.00	1.47	1.42	2.01	2.02	1.53	1.51	**1.29**
	Food	PSNR	35.28	25.71	33.25	39.96	43.42	38.46	36.43	40.67	41.59	**44.33**
		SSIM	0.919	0.643	0.946	0.978	0.989	0.986	0.983	0.985	0.984	**0.991**
		SAM	19.78	39.04	8.96	7.15	4.59	5.92	5.66	5.59	4.83	**4.23**
	Lemon	PSNR	36.67	38.33	34.96	38.61	44.45	33.26	33.09	42.41	42.72	**45.17**
		SSIM	0.941	0.958	0.959	0.977	0.991	0.971	0.968	0.988	0.988	**0.992**
		SAM	15.11	17.67	8.72	8.65	5.29	7.97	6.72	5.95	5.95	**4.91**
	Flowers	PSNR	38.15	16.85	34.40	40.23	45.39	41.87	39.31	43.07	41.56	**46.41**
		SSIM	0.922	0.460	0.948	0.975	0.991	0.989	0.986	0.986	0.983	**0.994**
		SAM	23.87	60.27	12.21	8.41	5.47	6.24	5.89	7.70	6.38	**4.29**
pavia		PSNR	38.97	41.14	32.14	41.08	42.75	42.03	39.82	37.45	40.57	**44.05**
		SSIM	0.984	0.985	0.953	0.988	0.992	0.990	0.987	0.977	0.987	**0.993**
		SAM	2.44	2.26	4.82	1.88	1.77	1.96	2.48	3.67	1.84	**1.55**

Fixing variable Θ, $\mathcal{S}$ subproblem is formulated as

$$\min_{\mathcal{S}} \|\mathcal{Y} - \mathbf{T}_\Theta(\mathcal{G}, \mathcal{E}) - \mathcal{S}\|_F^2 + \lambda\|\mathcal{S}\|_1. \tag{11}$$

Using the soft threshold operator, we have

$$\mathcal{S} = \text{Shrinkage}(\mathcal{Y} - \mathbf{T}_\Theta(\mathcal{G}, \mathcal{E}), \lambda/2) \tag{12}$$

where $\text{Shrinkage}(u, v) = \text{sign}(u)\max(|u| - v, 0)$.

It is worth noting that our proposed robust unsupervised learning strategy is as follows: the network updates the $\mathcal{S}$ subproblem (12) once again after every 20 iterations through the loss function (10), iteratively alternating in this manner until convergence.

IV. EXPERIMENTS

In this section, we present simulated and real experiments for two HSI quality improvement tasks: HSI fusion and denoising. The proposed UFFGNet is implemented in the PyTorch framework. We set the parameters λ and learning rate as 0.1 and 0.02 in all the experiments, respectively. The experimental results are obtained using an Intel i9-12900K processor alongside an NVIDIA GeForce RTX 3090 GPU.

A. Experiments on HSI Fusion

1) Simulated Dataset: Two publicly available datasets, the Columbia Computer Vision Laboratory (CAVE) dataset[1] and the University of Pavia (Pavia) dataset,[2] are selected as simulated data. The CAVE dataset comprises 32 high-quality indoor HSIs, each with a spatial resolution of 512 × 512. Furthermore, every HSI encompasses 31 spectral bands. Eight different HSI scenes (Balloons, Beads, Cd, Toy, Beers, Food, Lemon, and Flowers) are selected as simulated HR-HSI. Following the experimental setup of [52], we generate LR-HSI by averaging 32 × 32 pixel blocks and generate HR-MSI using the Nikon D700 camera's spectral response matrix. The Pavia dataset is collected by the reflective optics system imaging spectrometer (ROSIS), containing 610 × 340 spatial pixels and 103 spectral bands. After discarding the bands with low SNR and cropping the subregion, we select the upper left 256 × 256 spatial pixels containing 93 spectral bands as the HR-HSI. Following the experimental setup of [43], we use spectral response function from an IKONOS-like to simulate HR-MSI, and the LR-HSI is generated by averaging 8 × 8 pixel blocks.

2) Comparison Methods: We compare our proposed method with nine state-of-the-art HSI fusion approaches, including four traditional model-based methods HySure [36], FSTRD [43], GTNN [7], and CTDF [8], and two DL methods, UMC2FF [50], ADASR [32], and three guidance-based methods, GDD [52], CS2DIP [54], and uHNTC [51].

3) Evaluation Indices: Three quantitative indices, including peak-signal-to-noise-ratio (PSNR), structural similarity (SSIM), and spectral angle map (SAM), are used to evaluate the quality of restoration results. PSNR and SSIM are used to evaluate the spatial quality, whereas SAM is used to assess the spectral quality. Higher PSNR and SSIM values and lower SAM values indicate superior performance.

4) Results on Simulated Experiments: Table I presents a quantitative comparison of various methods in the simulated HSI fusion experiments. It can be observed that the proposed UFFGNet outperforms all other competing methods across all the evaluation metrics and scenes, achieving the highest level of performance. The visual comparison of the CAVE-Beers and Pavia datasets is presented in Fig. 3. It is evident that the proposed UFFGNet demonstrates superior fusion performance

[1]http://www1.cs.columbia.edu/CAVE/databases/multispectral

[2]https://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

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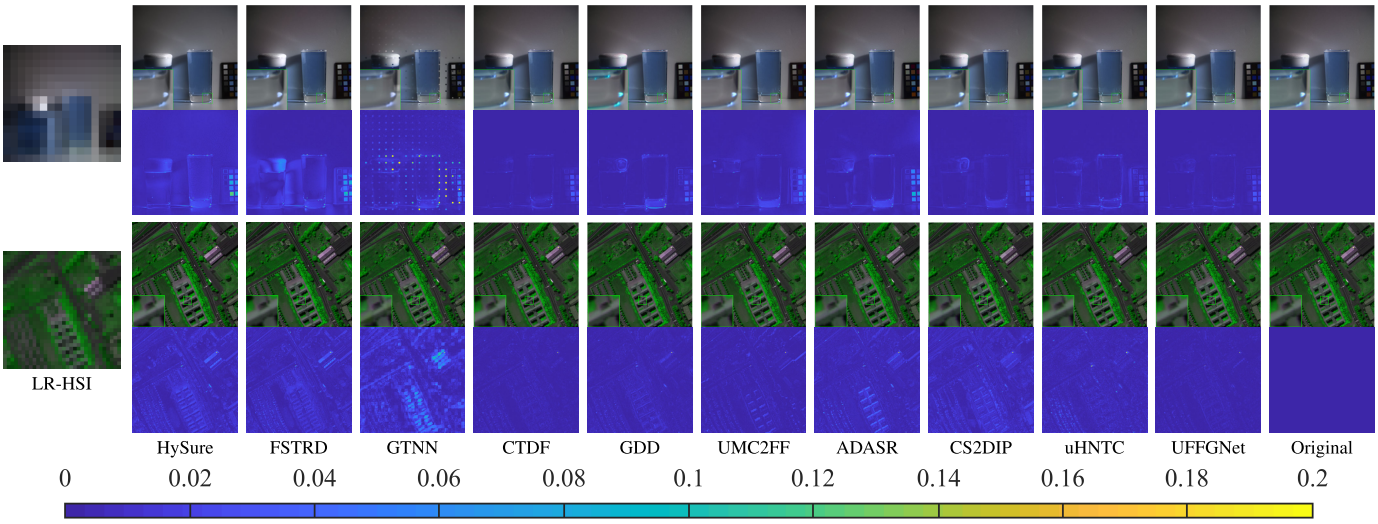


Fig. 3. Visual comparison and corresponding residual map of the simulated HSI fusion on the CAVE-Beers and Pavia datasets.

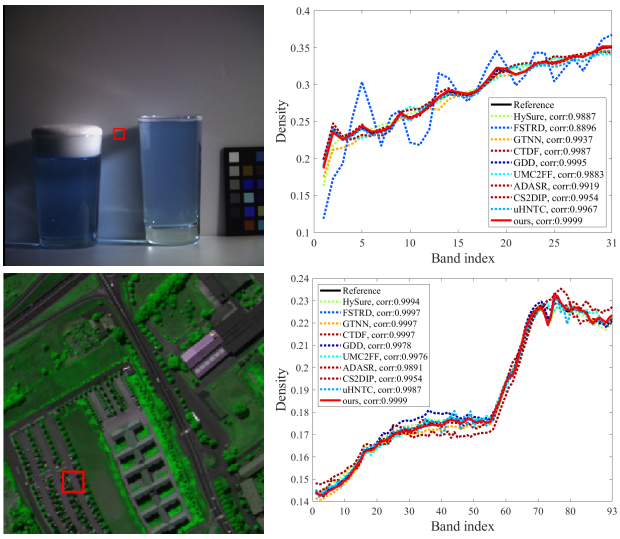


Fig. 4. (Top) Spectral curves of different methods on the CAVE-Beers image and (Bottom) Pavia image.

in both scenes, notably in preserving texture details more faithfully and achieving a smaller error map.

We further present the spectral curves to evaluate the spectral preservation capabilities of different methods. Fig. 4 shows the average spectral curves of a specific region in the Cave-Beers and Pavia scenes. It is observed that the spectral curve of the proposed method is closer to the original image than other competing methods, indicating the superiority of the proposed method in spectral fidelity.

5) Results on Real Experiments: Furthermore, we evaluate all the competing methods in the real HSI fusion experiment. The LR-HSI and HR-MSI are collected by Hyperion sensor with size $120 \times 120 \times 89$ and Sentinel-2A satellite with size $360 \times 360 \times 13$, respectively. We use four bands with a spatial resolution of 10 m for testing, resulting in an HR-MSI size of $360 \times 360 \times 4$. These four bands are derived from bands 2–4, and 8, corresponding to central wavelengths of 490, 560, 665, and 842 nm, respectively.

The visual comparison of all the comparison methods is provided in Fig. 5. It can be observed that all the methods enhance the spatial resolution. However, the results obtained by HySure, FSTRD, and UMC2FF perform excessive smoothness, and ADASR destroys the local texture information. In contrast, the proposed method is more capable of achieving perceptually pleasing and sharp images.

B. Experiments on HSI Denoising

1) Simulated Dataset: To further evaluate the robustness of the proposed UFFGNet, we conduct the experiments on HSI mixed noise removal. The simulated datasets used in this study include the hyperspectral Pavia City Center (PaC) dataset[3] and the Houston dataset.[4] The PaC dataset, with dimensions of $200 \times 200 \times 80$, is characterized by its complex structures and textures. For the Houston dataset, originally with dimensions of $349 \times 1905 \times 144$, the central region was extracted, and noisy bands were removed according to the procedure described in [34], resulting in a refined dataset with final dimensions of $256 \times 256 \times 124$.

To simulate complex noise scenarios, five different types of noise cases are added to the original HSI.

Case 1 (Gaussian Noise): Zero-mean Gaussian noise with a noise variance of 0.15 is added to all the bands.

Case 2 (Gaussian Noise + Impulse Noise): The Gaussian noise is added in the same manner as in Case 1, and the impulse noise is randomly sampled within the range of $[0, 0.2]$.

Case 3 (Gaussian Noise + Impulse Noise + Deadlines in 20% of Bands + Stripes in 20% of Bands): The Gaussian and impulse noise are the same as in Case 2. Moreover, we, respectively, choose 20% of all the bands to add deadlines and stripes, with the number varying from 5 to 12.

Case 4 (Gaussian Noise + Impulse Noise + Deadlines in 40% of Bands + Stripes in 40% of Bands): The Gaussian

[3]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

[4]https://hyperspectral.ee.uh.edu/?page_id=459

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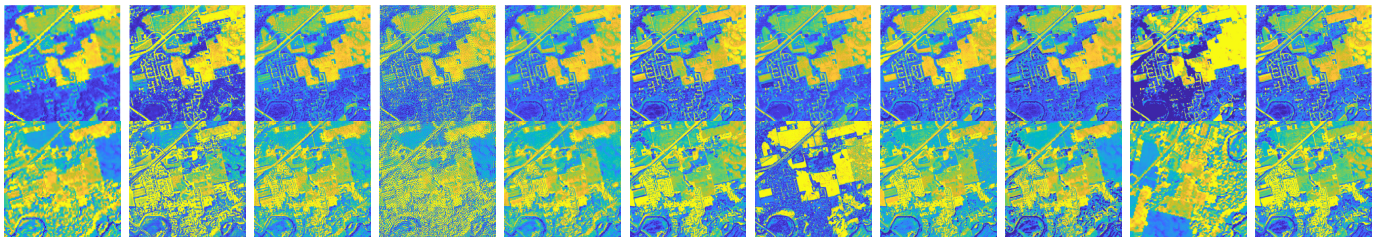


Fig. 5. Visual comparison of the real HSI fusion dataset. First row: the 2nd band of the fused HSI; second row: the 37th band of the fused HSI.

TABLE II
COMPARATIVE QUANTITATIVE ANALYSIS OF VARIOUS ALGORITHMS IN SIMULATED HSI DENOISING EXPERIMENTS

Dataset	Cases	Index	Noisy	FastHyMix [22]	RoSEGS [6]	ETPTV [15]	CNLRMA [9]	GDD [52]	DS2DP [66]	SST [27]	RCILD [23]	DLD [33]	UFFGNet
PaC	Case1	PSNR	18.25	42.21	42.68	35.03	32.95	40.91	34.80	36.03	43.04	37.66	**43.08**
		SSIM	0.456	0.994	0.994	0.966	0.946	0.992	0.954	0.989	0.994	0.976	**0.995**
		SAM	35.69	2.92	2.82	5.25	6.30	2.97	5.72	5.01	2.64	3.48	**2.60**
	Case2	PSNR	13.41	27.07	36.95	33.51	29.69	36.39	32.52	35.02	26.11	36.68	**40.32**
		SSIM	0.270	0.933	0.975	0.951	0.892	0.976	0.919	0.983	0.918	0.971	**0.990**
		SAM	42.55	8.73	4.21	5.78	8.36	3.65	6.45	5.54	8.78	3.69	**3.21**
	Case3	PSNR	13.48	27.80	35.38	33.59	29.28	37.95	33.04	35.02	26.34	36.85	**39.97**
		SSIM	0.265	0.933	0.968	0.950	0.885	0.982	0.931	0.983	0.918	0.973	**0.988**
		SAM	42.63	8.16	5.49	5.82	8.96	3.82	6.39	5.54	8.54	3.77	**3.37**
	Case4	PSNR	13.41	27.51	32.88	33.19	28.59	37.52	32.12	35.02	26.43	35.48	**39.47**
		SSIM	0.267	0.923	0.950	0.950	0.877	0.982	0.921	0.983	0.913	0.967	**0.989**
		SAM	43.19	8.99	7.25	6.01	9.55	3.91	7.05	5.54	9.01	4.48	**3.50**
	Case5	PSNR	13.36	27.23	32.10	32.95	27.84	36.22	31.70	35.02	26.39	35.94	**39.15**
		SSIM	0.263	0.915	0.939	0.946	0.861	0.979	0.915	0.983	0.908	0.969	**0.989**
		SAM	43.55	9.14	7.83	6.30	27.84	4.45	7.68	5.54	9.14	4.34	**3.69**
	Average	PSNR	14.38	30.36	36.00	33.65	29.67	37.80	32.84	35.22	29.66	36.52	**40.40**
		SSIM	0.304	0.940	0.965	0.953	0.892	0.982	0.928	0.984	0.930	0.971	**0.990**
		SAM	41.52	7.59	5.52	5.83	12.20	3.76	6.66	5.44	7.62	3.95	**3.27**
Houston	Case1	PSNR	16.84	39.88	40.12	35.60	33.64	38.97	34.19	33.64	42.19	36.24	**42.35**
		SSIM	0.180	0.973	0.971	0.909	0.852	0.974	0.874	0.956	0.982	0.924	**0.989**
		SAM	55.39	4.56	4.18	6.76	8.73	4.07	7.45	10.94	3.48	5.16	**2.85**
	Case2	PSNR	13.15	28.44	33.18	35.45	27.60	37.33	34.52	31.34	23.58	35.58	**39.46**
		SSIM	0.145	0.888	0.892	0.908	0.624	0.953	0.885	0.897	0.791	0.912	**0.969**
		SAM	54.51	14.81	8.49	6.76	22.33	5.77	6.27	14.28	16.76	5.16	**4.87**
	Case3	PSNR	12.65	27.11	32.41	35.46	26.83	37.07	34.21	31.34	23.42	36.20	**39.12**
		SSIM	0.137	0.877	0.858	0.908	0.590	0.953	0.881	0.897	0.784	0.925	**0.968**
		SAM	55.26	17.83	9.73	6.47	24.27	6.11	7.03	14.28	16.52	**4.72**	5.19
	Case4	PSNR	12.79	28.02	32.45	35.35	26.90	36.05	33.29	31.34	23.58	35.71	**38.92**
		SSIM	0.136	0.886	0.862	0.907	0.603	0.929	0.858	0.897	0.789	0.907	**0.966**
		SAM	55.33	16.70	9.91	6.91	24.14	6.44	8.65	14.28	16.43	**5.14**	5.49
	Case5	PSNR	12.82	26.89	32.53	35.03	26.32	37.16	33.21	31.34	23.34	35.66	**38.80**
		SSIM	0.138	0.860	0.863	0.899	0.586	0.957	0.861	0.897	0.786	0.917	**0.964**
		SAM	55.56	18.74	9.72	6.71	25.77	6.37	8.21	14.28	16.56	**5.49**	5.58
	Average	PSNR	13.65	30.07	34.14	35.38	28.26	37.32	33.88	31.80	27.22	35.88	**39.73**
		SSIM	0.147	0.897	0.889	0.906	0.651	0.953	0.872	0.909	0.827	0.917	**0.971**
		SAM	55.21	14.53	8.41	6.72	21.05	5.75	7.52	13.61	13.95	5.13	**4.79**

noise and impulse noise are the same as in Case 2. Specifically, we, respectively, select 40% of all the bands to add deadlines and stripes, with the number varying from 5 to 12.

Case 5 (Gaussian Noise + Impulse Noise + Deadlines in 60% of Bands + Stripes in 60% of Bands): The Gaussian noise and impulse noise are the same as in Case 2. Specifically, we select 60% of all the bands separately to add deadlines and stripes, with the number varying from 5 to 12.

Since real noisy HSIs contain many HSNRBs [67], ten bands are randomly selected from the simulation data to replace with the clean bands, and these ten bands are used as the guidance image. It is worth noting that these clean bands are also used in the comparison methods to ensure fairness.

2) Comparison Methods: We compare our proposed UFFGNet with nine state-of-the-art methods, including three traditional model-based methods, RoSEGS [6], ETPTV [15] and CNLRMA, [9], and five DL methods FastHyMix [22], DS2DP [66], SST [27], DLD [33], and RCILD [23] and one guidance-based method GDD [52]. We also use PSNR, SSIM, and SAM metrics to evaluate denoising performance.

3) Results on Simulated Experiments: Three quantitative assessment results for the PaC and Houston datasets are presented in Table II. It can be observed that our method achieves the best performance across all the cases for the PaC dataset, while also demonstrating superior performance on the Houston dataset. Especially when the HSI suffers from mixed noise, the proposed method is superior to other compared methods. Fig. 6 provides the visual comparison and residual results for the PaC dataset under Case 5 and the Houston dataset under Case 3. It is evident that the proposed method not only effectively removes noise but also preserves spatial details. In contrast, the comparison methods, such as FastHyMix, RoSEGS, CNLRMA, and RCILR, exhibit notable residual stripes, while GDD, DS2DP, and SST show spectral distortion. Moreover, our method produces a minimum error, further validating the superiority of our method for HSI denoising.

4) Results on Real Experiments: To further validate the generalization of our method, we implement the experiment on a real-world noisy HSI dataset. The GF-5 dataset consists of 256 × 256 pixels with 155 bands, which is mainly

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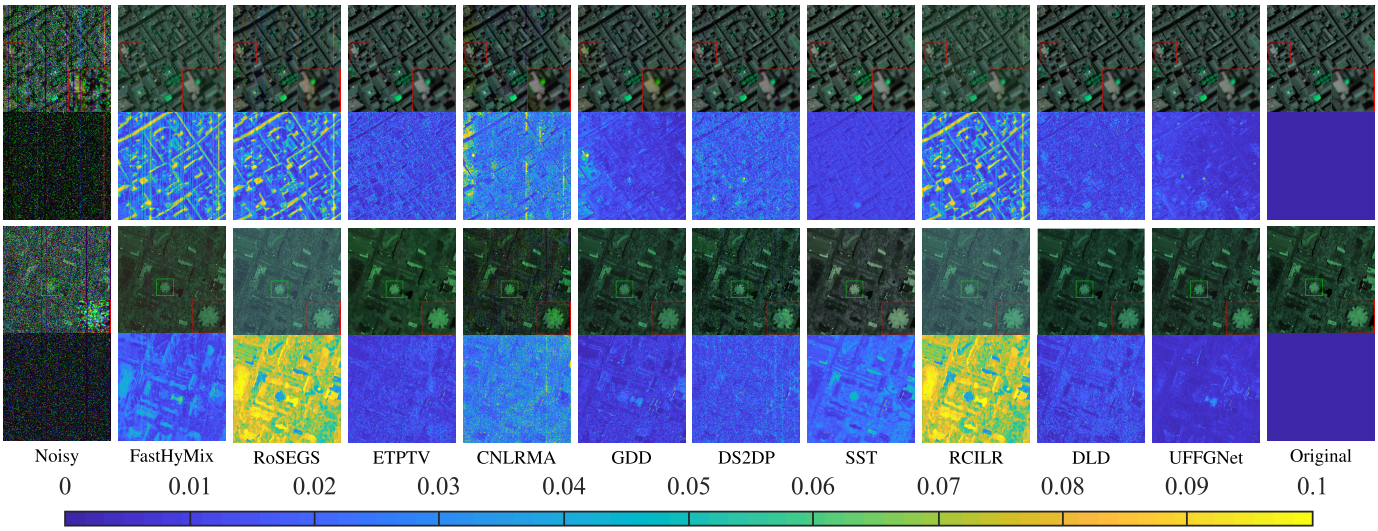


Fig. 6. Visual comparison and corresponding residual map of the simulated HSI denoising on the PaC dataset under Case 5 and Houston under Case 3.

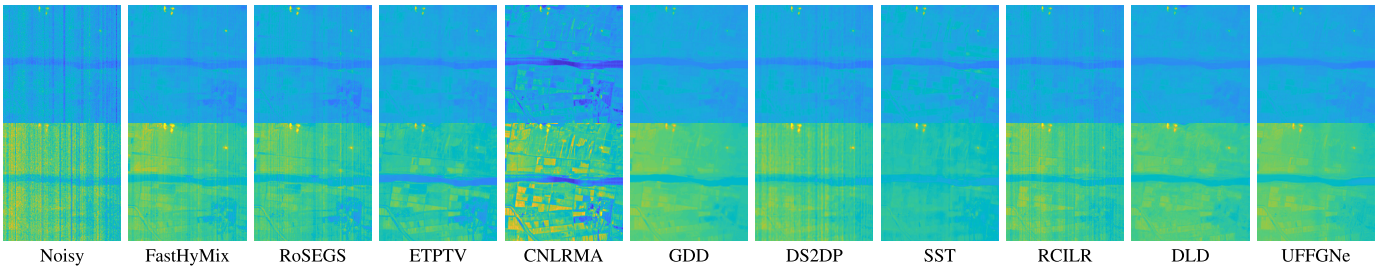


Fig. 7. Visual comparison of the real GF-5 dataset. First row: the 153rd band of the denoising result; second row: the 155th band of the denoising result.

polluted by Gaussian noise, impulse noise, and stripe noise. Since our method and the GDD method require guidance images, we select nine relatively clean bands as guidance images by visual comparison. Due to the unavailability of ground truth for real datasets, it is not feasible to use evaluation metrics to assess the effectiveness of various methods. The denoising visual results are presented in Fig. 7. The proposed approach outperforms the comparison methods in eliminating mixed noise and retaining more detailed texture information.

C. Ablation Study and Discussion

To demonstrate the effectiveness of the proposed UFFGNet, the role of each module of the proposed UFFGNet is analyzed. The ablation studies are conducted on the simulated HSI denoising task using the PaC dataset.

1) Effectiveness of the FAM: To fully use the encoded and decoded features, we design the FAM to mitigate information loss during the encoding–decoding process. To verify the effectiveness of the FAM, we use common summation and concatenation operations to fuse the encoded and decoded features. Table III lists the PSNR, SSIM, and SAM values for the denoising task. We observe that the FAM significantly improved denoising performance compared with the summation and concatenation operations.

2) Effectiveness of the SFEM: The SFEM is designed to squeeze the channel features, effectively exploiting the channel dependency. Table III lists the denoising results of UFFGNet without the SFEM (UFFGNet w/o SFEM). We observe that leveraging the SFEM can boost denoising performance, verifying its effectiveness in extracting spectral dependencies. To further analyze the interaction between FAM and SFEM, we evaluate UFFGNet without both the modules [UFFGNet w/o (FAM & SFEM)]. As presented in Table III, the absence of these modules leads to a notable decline in performance. Furthermore, comparisons show that FAM and SFEM mutually reinforce each other, enhancing denoising performance by effectively using both the spatial and spectral features.

3) Effectiveness of the DFE Network: DFE network is effective in extracting multiscale spatial–spectral features from guidance images. In contrast, without the guidance information

TABLE III
AVERAGE RESULTS OF THE ABLATION STUDY ACROSS FIVE CASES ON THE PaC DATASET

Method	FAM	SFEM	DFE	Robust	PSNR	SSIM	SAM
FAM → Summation	✗	✓	✓	✓	39.16	0.987	3.60
FAM → Concatenation					39.41	0.988	3.44
UFFGNet w/o SFEM	✓	✗	✓	✓	39.38	0.989	3.35
UFFGNet w/o DFE	✓	✓	✗	✓	33.09	0.936	5.05
UFFGnet w/o Robust	✓	✓	✓	✗	27.91	0.861	10.08
UFFGNet w/o (FAM & SFEM)	✗	✗	✓	✓	37.24	0.985	3.56
UFFGNet	✓	✓	✓	✓	**40.40**	**0.990**	**3.27**

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TABLE IV
NETWORK PARAMETERS, AVERAGE EXECUTION TIME, AND DENOISING PERFORMANCE OF ALL METHODS EVALUATED ON THE SIMULATED PaC DATASET

Index	Noisy	FastHyMix [22]	RoSEGS [6]	ETPTV [15]	CNLRMA [9]	GDD [52]	DS2DP [66]	SST [27]	RCILD [23]	DLD [33]	UFFGNet
Parameters (M)	-	0.49	**0**	**0**	**0**	1.38	1.17	4.14	2.89	1.38	1.48
Time (s)	-	5.44	54.71	21.27	53.73	30.55	272.28	**1.23**	4.64	1051.94	34.60
PSNR	14.38	30.36	36.00	33.65	29.67	37.80	32.84	35.22	29.66	36.52	**40.40**
SSIM	0.304	0.940	0.965	0.953	0.892	0.982	0.928	0.984	0.930	0.971	**0.990**
SAM	41.52	7.59	5.52	5.83	12.20	3.76	6.66	5.44	7.62	3.95	**3.27**

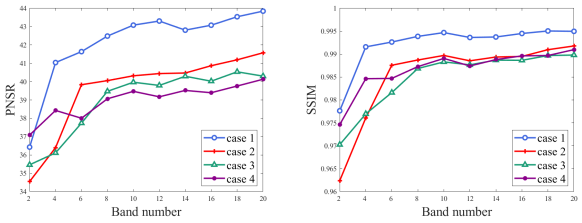


Fig. 8. Sensitivity analysis of the number of HSNRBs on the PaC dataset.

TABLE V
COMPREHENSIVE QUANTITATIVE COMPARISON WITH HIR-Diff ON GAUSSIAN NOISE REMOVAL

Standard deviation		30		50		70	
Dataset	Method	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
WDC mall	HIR-Diff [34]	42.85	0.97	40.77	0.94	39.33	0.92
	UFFGNet	**44.69**	**0.99**	**41.24**	**0.98**	**40.85**	**0.97**
Houston	HIR-Diff [34]	38.13	0.94	36.01	0.90	34.56	0.86
	UFFGNet	**42.98**	**0.99**	**41.09**	**0.99**	**39.35**	**0.98**
Salinas	HIR-Diff [34]	43.79	0.96	**41.95**	0.95	39.48	0.93
	UFFGNet	**44.79**	**0.98**	41.30	**0.96**	**39.85**	**0.95**

from the DFE network, the AFG network would be forced to rely solely on random noisy input, which typically lacks the spatial and spectral fidelity that a guidance image provides. The result of removing the DFE network (UFFGNet w/o DFE) is presented in Table III. The notable decrease in denoising effectiveness highlights the crucial role of guidance images in image restoration.

4) Effectiveness of Robust Unsupervised Network: To validate that the proposed UFFGNet can handle sparse components, we extend the UFFGNet to robust unsupervised learning. As shown in Table III, UFFGNet without robust unsupervised learning (UFFGNet w/o Robust) experiences a significant performance decline. By incorporating sparse errors into the learning process, the model improves its adaptability to different types of noise, resulting in significantly enhanced HSI denoising performance in real-world environments.

5) Network Parameters and Execution Time: Table IV shows comparison of network parameters, average execution times, and denoising performance for all the evaluated methods on the simulated PaC dataset. While the supervised SST method demonstrates excellent computational efficiency during testing, its dependence on large training datasets and limited generalization reduces its practicality. Similarly, RCILD achieves rapid execution via pretrained denoisers but struggles with complex mixed noise. The proposed method, while requiring slightly more time, excels in denoising performance and maintains a compact network design, emphasizing efficiency without compromising quality. Overall, the proposed method achieves superior performance with an optimal tradeoff between computational cost and complexity, making it a highly effective solution for HSI denoising tasks.

6) Number of HSNRBs' Analysis: In our proposed method, HSNRBs are used as guidance images. To evaluate the impact of the number of HSNRB, we conduct experiments on the simulated PaC dataset under various noise cases. Fig. 8 presents the PSNR and SSIM results under different numbers of HSNRB. It can be clearly observed that as the number of HSNRB increases, the denoising performance improves, but it stabilizes within a certain range. To balance denoising performance with practical application considerations, we fix the number of HSNRB at ten in all the experiments.

7) Comparison to Diffusion Model-Based HIR-Diff: Recently, diffusion models have garnered significant attention. To validate the competitive performance of the proposed method against diffusion models, we compare it with HIR-Diff [34], a state-of-the-art approach based on diffusion techniques. Since HIR-Diff is designed for Gaussian noise removal, it performs poorly on mixed noise. Therefore, we compare our method with HIR-Diff under Gaussian noise conditions. To ensure a fair comparison, we use the Washington DC Mall, Houston, and Salinas datasets from HIR-Diff with identical Gaussian noise degradation. As shown in Table V, our method outperforms HIR-Diff in all the datasets and noise intensities, except for the case with noise level 50 in the Salinas dataset, where the PSNR is lower than that of HIR-Diff.

V. CONCLUSION

In this work, we propose UFFGNet, a unified framework with sparse prior constraints for unsupervised HSI fusion and denoising, aiming to enhance HSI quality across diverse datasets. UFFGNet is a network architecture-based regularizer and can be used to learn the implicit deep prior directly from the observed image. The proposed method eliminates the need for training data and avoids reliance on handcrafted priors, thereby achieving strong generalization capability. To mitigate the influence of sparse components in the observed image on the network, we extend the proposed UFFGNet to robust unsupervised network learning. Extensive experiments further demonstrate that UFFGNet can be effectively applied to HSI fusion and denoising tasks, achieving state-of-the-art performance across various datasets. It alleviates the limitations of most studies, which focus on solving only one specific problem, require a large number of training datasets, and are sensitive to variations in noise distribution.

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